Hours of Sleep	Exam score (%)
3	55
3.5	57
4	58
4	65
4.5	70
11	72
5	73
11.5	74
7	77
6	79
5.5	80
7.5	82
10	83
7	84
10.5	85
6.5	88
8	88
6.5	90
9.5	90
7.5	91
8	93
7	94
8.5	96
9	97
8.5	100

Hours of Sleep	Exam Score				Average Hours of sleep:	
5	84					
6	88					
4	79					
6	90				Median Exam Score:	
7	85					
8	92					
12	95					
2	69					
7	81				Standard Dev:	
5	64					
3	61					
4	68					
8	90					
9	92					
3	57					
2	52					
4	59					
6	60					
7	65					
5	72					
8	88					
3	61					
6	80					
11	90					
4	73					

Hours of Sleep	Exam score (%)		Regression Measure	Calculated Result		Regression Statistics			Slope Significance Test	
			Slope						t-statistic	
			Intercept						Residual degrees of freedom	
			R-squared						Two-tailed p-value	

Reflection Questions
Step 3.2: Pause and predict: based on your scatterplot, do you expect the relationship between sleep and exam scores to be positive, negative, weak, or strong? Record your prediction before moving on. Don't worry about getting it right just yet, we're about to put this data to work.
Step 5.2: Slope interpretation:
Step 5.2: Intercept interpretation:
Step 5.2: Use the simple linear regression equation y=b0+b1x to predict the exam score for a student who gets 7 hours of sleep
Step 6.1: What is the R^2 value for your linear regression model (cell E4), and what does this value tell you about how well hours of sleep explains the variation in exam scores?
Step 6.3: What is the p-value for your linear regression model? Using the significance level, do you reject or fail to reject the null hypothesis? What does your decision tell you about the evidence for a linear relationship between hours of sleep and exam scores?
Step 7: Based on your scatterplot and the statistics calculated regarding it, does a straight line model appear to capture the overall pattern in the data? Why or why not?

Hours of Sleep	Exam score (%)		Regression Measure	Regression Values		Regression Statistics	
			Slope				
			Intercept				
			R-squared				

Reflection Questions

Step 8: Looking at your scatterplot and linear trendline, does a straight line appear to describe the relationship between hours of sleep and exam scores well? Why or why not?

What pattern do you notice in the data that a straight line might fail to capture?

Based on your scatterplot, why might a quadratic model be more appropriate than a linear model?

Step 9: What does squaring the hours of sleep allow us to do when modeling the relationship between sleep and exam scores?

If a student sleeps 7 hours, what is their squared sleep-hours value? Show your calculation

Step 11: What are the 3 coefficients in your quadratic regression equation, and what does each coefficient represent?

Is the coefficient of the squared term positive or negative? Why does its sign suggest the shape of the curve?

Step 12: What is the R-squared value for your linear model? What is the R-sqaured

Which model has the higher R-squared, and what does this suggest about how well each model fits your dataset?

Why isnt a higher R-squared value alone enough to conclude that the quadratic model is better?

Final takeaway: Based on your analysis, what can you conclude about the relationship between hours of sleep and exam scores?

How does quadratic regression help us understand a relationship that linear regression might miss?

Does your analysis prove that getting a certain amount of sleep causes students to perform better on exams? Why or why not?